

EcoOrbit | methane and pollutant intelligence

- Three international CH₄ case-study sites; QA-screened satellite workflow
- Tanager land-use classification is a supporting feature
- Team: [add] | Official theme: [add]

Methane evidence question

- Stanford Arizona: verified 2022 controlled releases.
- Hassi R Mel: published regional methane study; Jeddah landfill target.
- Proposed Jeddah 2024 dates require satellite verification.

Pollutant data and provenance

- Sentinel-5P CH₄: real US regional observations included.
- New CH₄ case scenes pending; no claimed target plume.
- Carbon Mapper / EMIT CH₄ and CO₂ exports add rate, wind, uncertainty.

CH4 method

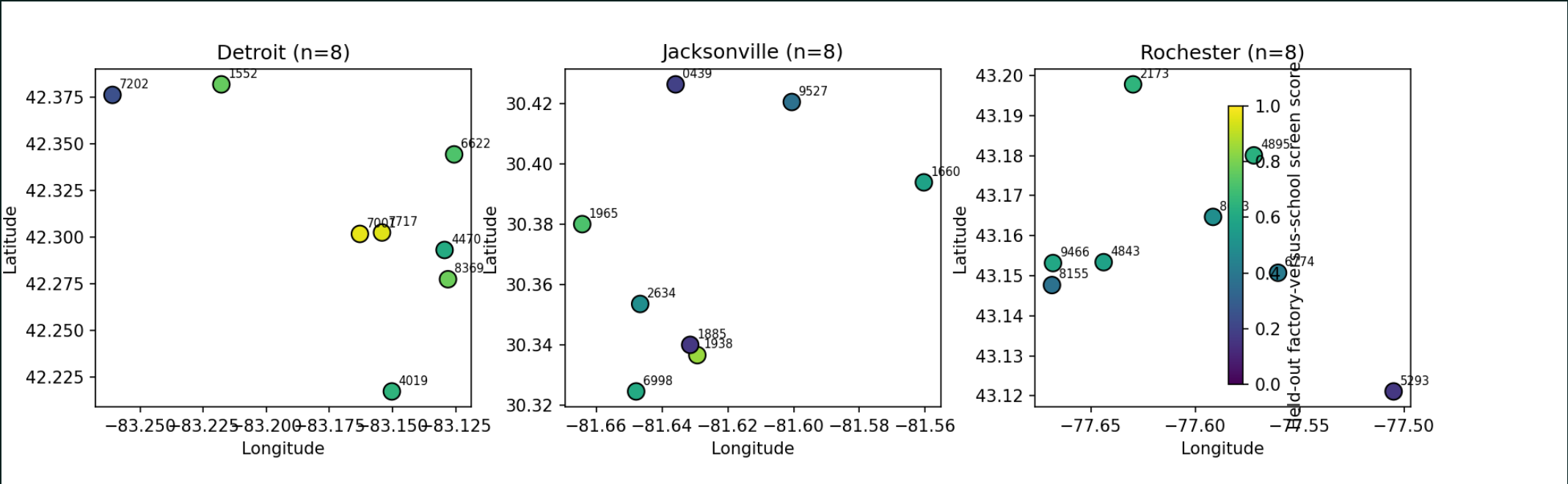
- TROPOMI XCH4 ppb: QA > 0.5 and 0-25 km versus 25-80 km.
- A regional column contrast cannot establish site emissions.
- Check hyperspectral plume image, wind and source geometry.

Real hyperspectral plumes

- Brazil landfill: Tanager CH₄, 2,836 kg/h (29 Sep 2024).
- India power generation: Tanager CO₂, 838,000 kg/h (1 Nov 2024).
- Carbon Mapper EMIT example: CH₄ 3,610.58 +/- 377.95 kg/h.

Published examples; the three proposed case targets still require observed scenes.

Supporting factory-site screen



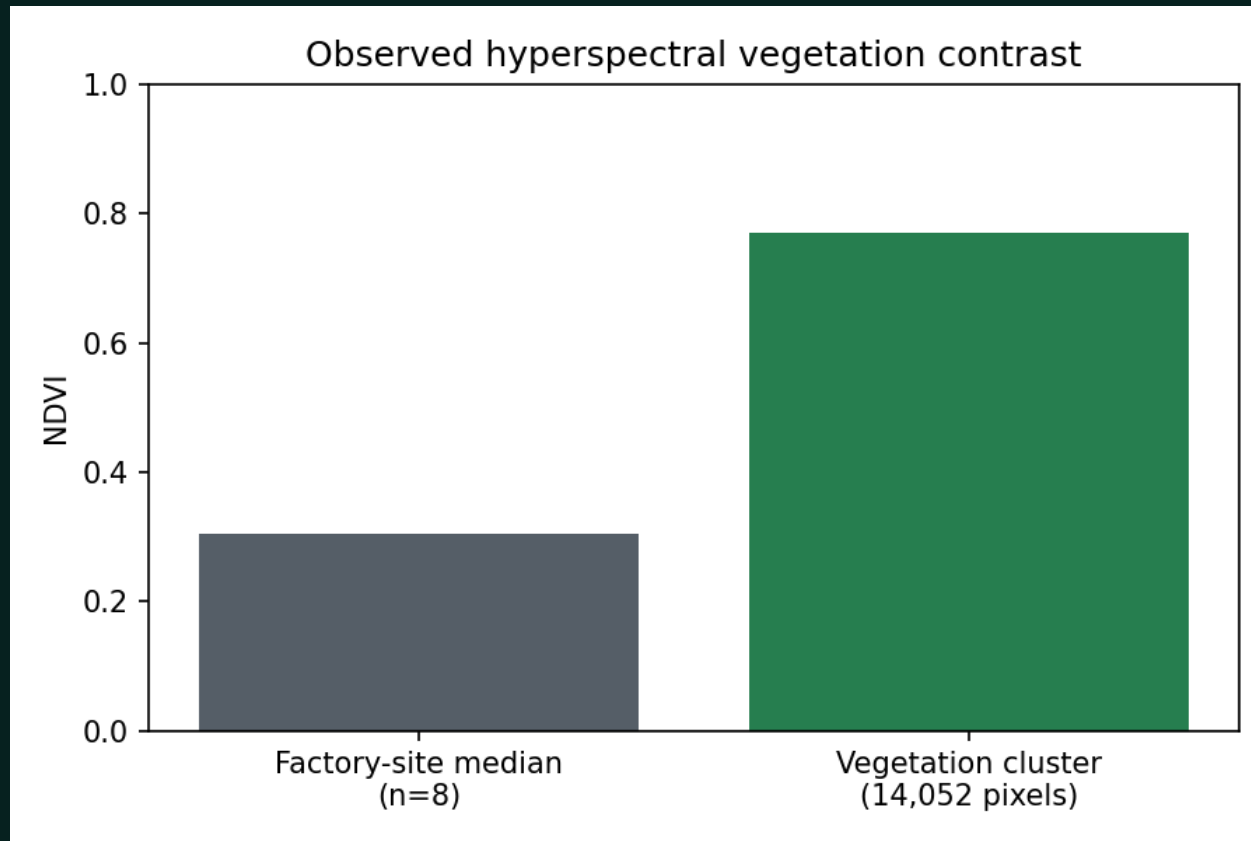
24 points across three US metros; point scores do not validate building polygons.

Held-out evaluation



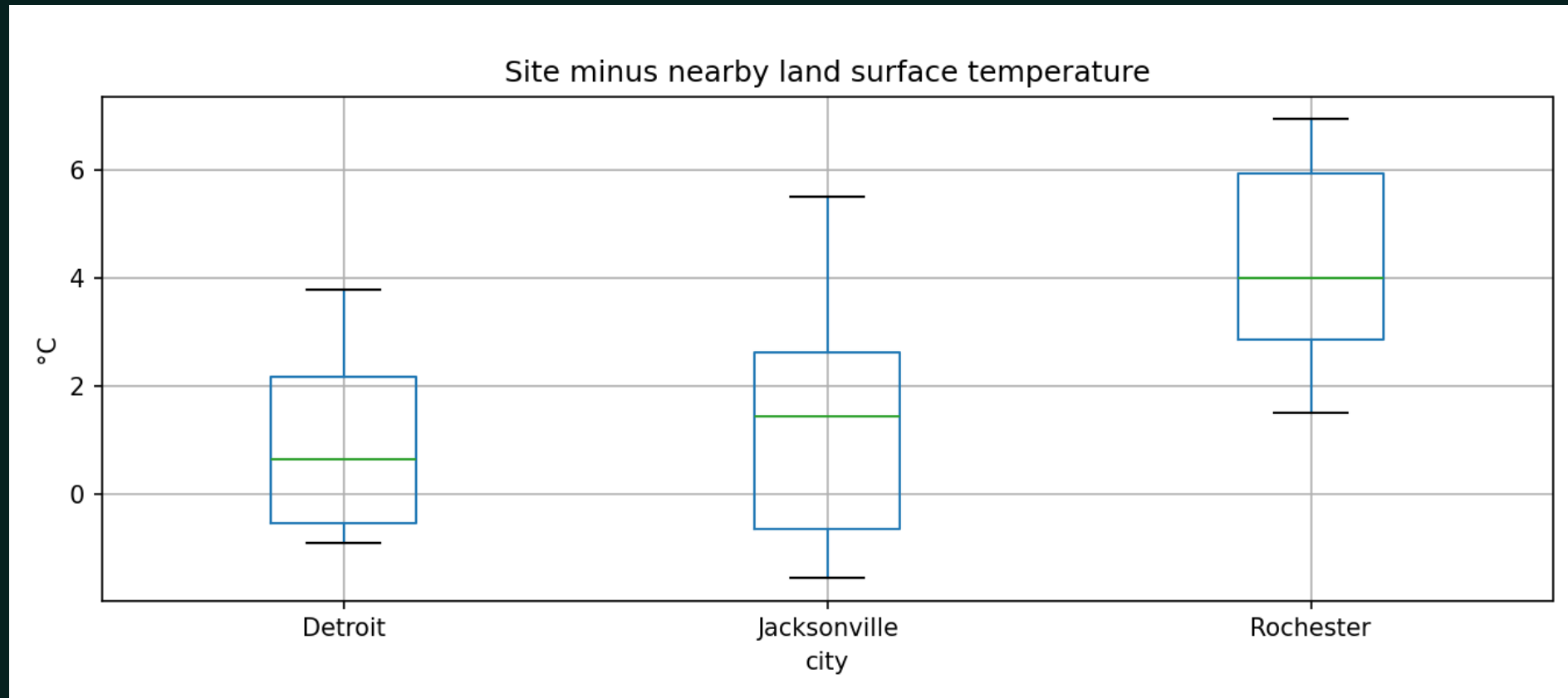
12 test sites across three cities: 11/12 accuracy, AUC 0.944; school-only controls.

Observed green reference



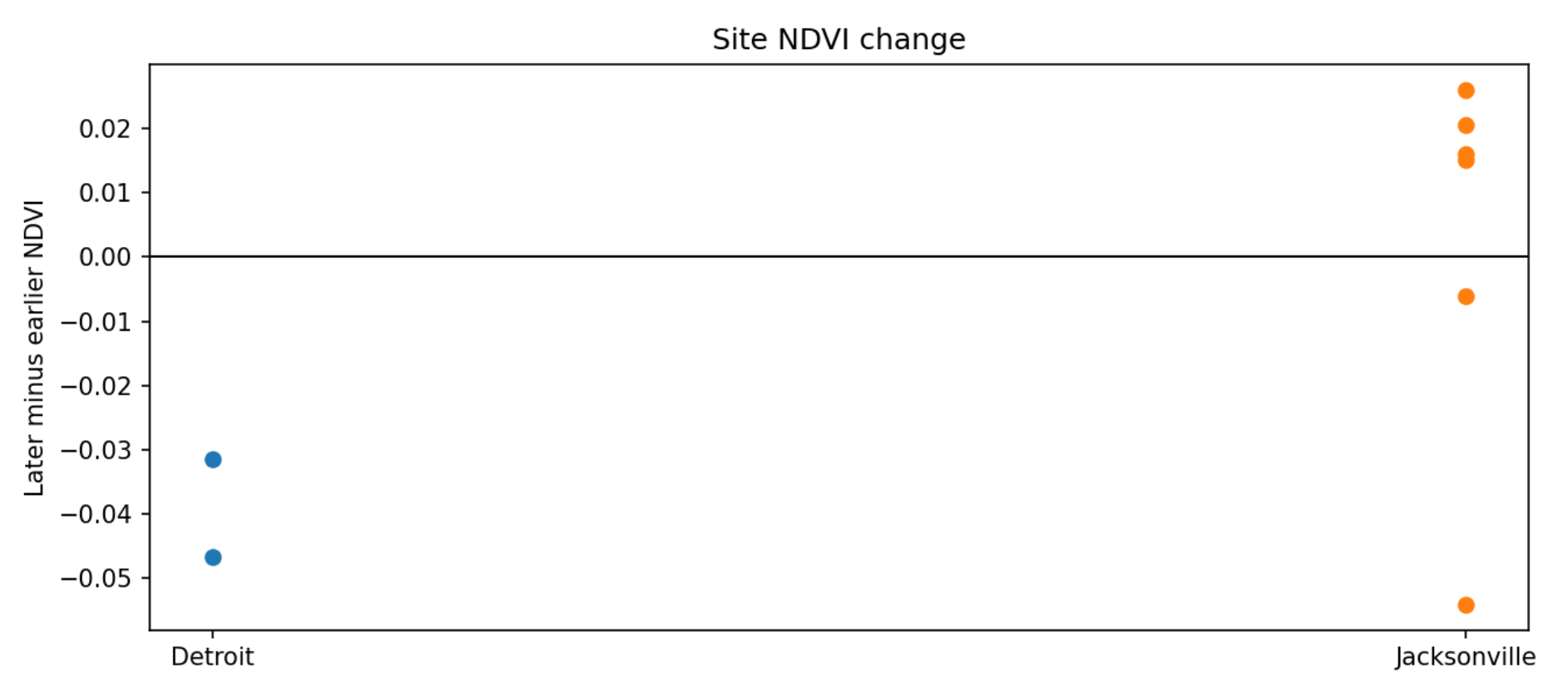
Jacksonville Tanager: green cluster NDVI 0.770 vs 8 factory-neighborhood median 0.305; not matched parcels.

Observed surface heat



19 distinct Landsat sites, 38 dated rows; surface contrast is not waste-heat flux.

Two-date Sentinel-2 change



Eight paired sites: six Jacksonville, two Detroit; no valid Rochester pixels in selected scenes.

Satellite pollutant columns

- TROPOMI NO₂ and SO₂: observed at eight Detroit sites.
- Multi-kilometre pixels may include multiple factories.
- Jacksonville download pending; Rochester no image-week items.

Carbon and interactive dashboard

- Gas, case and date filters; NO₂, SO₂, PM and aerosol selectors.
- Instantaneous CH₄/CO₂ plume kg/h is not an annual inventory.
- Facility activity and sourced factors enable Scope 1/2 tCO₂e.

ESG evidence and next steps

- GRI 305 / GHG Protocol gaps shown until records are supplied.
- Verify plume source, wind, operating period and site polygons.
- Landfill and factory classes require validated training examples.